

Research on AI-Based Garment Warping and Geometric Matching Techniques for Virtual Try-On Systems

AI/ML Internship — Day 9

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Introduction

Virtual Try-On (VTON) systems have become one of the most promising applications of Artificial Intelligence in the fashion industry. These systems allow users to digitally wear garments by combining computer vision, deep learning, and image synthesis technologies.

One of the most challenging components of a Virtual Try-On system is garment warping. Since every user has a different body shape and pose, the clothing image must be transformed and aligned accurately before generating the final output.

This research focuses on studying modern garment warping techniques, Geometric Matching Modules (GMM), Thin Plate Spline (TPS) transformation, HR-VITON architecture, and various methods used to improve realistic cloth fitting.

Objectives

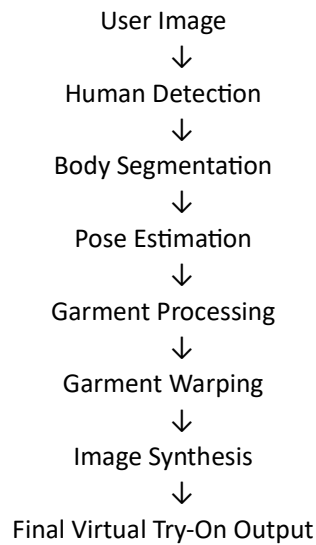
The objectives of this research are:

- Study AI-based garment warping techniques.
 - Understand Thin Plate Spline (TPS) transformation.
 - Analyse the Geometric Matching Module (GMM).
 - Explore HR-VITON architecture.
 - Study warp control points and affine transformations.
 - Understand common edge cases in Virtual Try-On systems.
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1. Virtual Try-On Systems

Virtual Try-On systems allow users to visualize garments on their own body without physically wearing them.

The overall workflow generally consists of:



Garment warping is one of the most important stages because it determines how naturally the clothing adapts to the user's body.

2. Garment Warping

Overview

Garment warping is the process of transforming a clothing image so that it matches the body shape and pose of the user.

Instead of simply overlaying a shirt on top of a person, AI systems stretch, rotate, and deform the garment while preserving its texture and appearance.

Importance of Garment Warping

Garment warping helps achieve:

- Better clothing alignment.
- Improved fitting quality.
- Natural fabric deformation.
- Better user experience.
- Realistic Virtual Try-On outputs.

Without proper warping, garments appear unrealistic and fail to match body movements.

3. Thin Plate Spline (TPS)

Overview

Thin Plate Spline (TPS) is a mathematical transformation technique widely used in image registration and Virtual Try-On systems.

TPS smoothly deforms one image to match another by using a set of control points.

Working Principle

TPS uses corresponding points between the garment image and the user's body.

Example control points:

- Left Shoulder
- Right Shoulder
- Left Waist
- Right Waist

The algorithm calculates a smooth transformation that minimizes distortion while preserving garment structure.

Advantages

- Smooth deformation.
 - Low computational cost.
 - Preserves garment texture.
 - Easy integration into AI pipelines.
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Limitations

- Performance decreases with extreme body poses.
 - Limited flexibility for highly complex clothing.
 - Requires accurate control point estimation.
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4. Geometric Matching Module (GMM)

Overview

The Geometric Matching Module is a deep learning network designed to estimate geometric transformations between garments and human bodies.

Instead of manually defining transformation rules, GMM learns how garments should be warped during training.

Working Process

The GMM pipeline consists of:

1. Feature Extraction
 2. Correlation Computation
 3. Transformation Parameter Estimation
 4. TPS Warping
 5. Warped Garment Generation
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Benefits

- High fitting accuracy.
 - Handles different body shapes.
 - Learns complex transformations automatically.
 - Produces realistic garment alignment.
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5. HR-VITON

Overview

HR-VITON stands for High-Resolution Virtual Try-On Network.

It is an advanced deep learning architecture developed to improve image quality and garment realism.

Key Features

- High-resolution image generation.
 - Better preservation of garment textures.
 - Improved sleeve and collar alignment.
 - Enhanced edge quality.
 - Better handling of complex body poses.
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Advantages over Traditional Models

Traditional Models	HR-VITON
Lower resolution	High resolution
Texture distortion	Better texture preservation
Simple alignment	Advanced geometric matching
Visible artifacts	Cleaner outputs

6. Warp Control Points

Control points define the regions used for garment transformation.

Common control points include:

- Left Shoulder
- Right Shoulder
- Chest Centre
- Left Waist
- Right Waist

These points guide TPS and GMM during the warping process.

Accurate control point estimation significantly improves Virtual Try-On quality.

7. Affine Transformation

Overview

Affine Transformation is a geometric image transformation technique commonly used in computer vision and image processing applications. In Virtual Try-On systems, it provides a simple method for adjusting garment images before applying more advanced warping algorithms.

Unlike complex deep learning models, affine transformation relies on mathematical operations to modify the position and orientation of an image while preserving straight lines and relative distances between points.

The primary operations performed by affine transformation include:

- Translation
- Rotation
- Scaling
- Shearing

These operations help align garments with the user's body during the initial stages of the Virtual Try-On pipeline.

Working Principle

Affine transformation maps the coordinates of one image onto another using a transformation matrix. The algorithm calculates how each pixel should be repositioned to achieve the desired orientation and scale.

In Virtual Try-On applications, affine transformation may be used to:

- Adjust garment size.
- Rotate clothing according to body posture.
- Shift garments to the correct body position.
- Correct small alignment errors.

Advantages

Affine transformation offers several benefits:

- Fast computation speed.
- Low memory requirements.
- Easy mathematical implementation.
- Stable and predictable performance.
- Suitable for preprocessing tasks.

Because of its simplicity, affine transformation is often used as an initial alignment step before more advanced geometric matching methods.

Limitations

Despite its usefulness, affine transformation has several limitations.

- It cannot accurately model complex cloth deformations.
- Fabric folds and wrinkles cannot be generated.
- It does not adapt well to highly dynamic body poses.
- Garment realism is limited compared to deep learning methods.

For this reason, modern Virtual Try-On systems typically combine affine transformation with TPS and deep learning-based warping networks.

8. Edge Cases in Garment Warping

Overview

Garment warping becomes increasingly difficult when the input image contains unusual body positions or environmental conditions. These special situations are commonly referred to as edge cases.

Handling edge cases is essential for producing stable and realistic Virtual Try-On results.

Extreme Body Poses

Users may upload images where their body is rotated or tilted significantly.

Examples include:

- Side-facing poses.
- Raised arms.
- Bent body positions.
- Sitting posture.

Such poses make garment alignment more challenging because body landmarks become difficult to estimate accurately.

Loose or Oversized Clothing

Oversized garments contain additional folds and wrinkles that cannot be easily modelled using simple geometric transformations.

AI systems must estimate:

- Fabric stretching.
- Natural folds.
- Sleeve movement.
- Hanging portions of clothing.

Advanced Virtual Try-On models use deep learning to approximate these deformations.

Occlusions

Occlusion occurs when one object partially blocks another.

Common examples include:

- Hands covering clothing.
- Bags crossing the torso.
- Long hair covering shoulders.
- Accessories hiding garment regions.

Occlusions reduce segmentation accuracy and complicate garment fitting.

Complex Backgrounds

Background objects sometimes interfere with body detection algorithms.

Challenges include:

- Similar clothing and background colours.
- Multiple people in the image.
- Furniture overlapping the user.
- Poor lighting conditions.

Modern AI systems attempt to reduce these problems through improved segmentation and background removal techniques.

Different Body Shapes

Virtual Try-On systems must support users with varying:

- Heights.
- Body proportions.
- Shoulder widths.
- Waist sizes.

The garment must adapt naturally to all these variations without losing its original appearance.

Handling Edge Cases

Modern AI architectures address these challenges through:

- Large-scale training datasets.
 - Better landmark detection.
 - Geometric Matching Modules.
 - TPS transformations.
 - Advanced image synthesis networks.
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9. Applications

AI-based garment warping and geometric matching techniques have numerous real-world applications.

Virtual Try-On Systems

Customers can digitally try garments before purchasing, reducing uncertainty during online shopping.

Smart Retail Platforms

Retail companies use AI fitting systems to improve customer engagement and reduce return rates.

AI Fashion Assistants

Virtual assistants recommend garments that match the user's body shape and preferences.

Online Clothing Stores

Fashion platforms integrate Virtual Try-On features to enhance the shopping experience.

Personalized Recommendation Systems

AI models combine body measurements with garment information to provide customized outfit suggestions.

Digital Fashion and Metaverse Applications

Virtual avatars and digital humans use garment warping technologies to wear realistic digital clothing.

Research and Academic Applications

Many universities and research organizations use Virtual Try-On datasets and warping models to develop new AI algorithms.

10. Research Findings

The study indicates that garment warping is one of the most important stages in the Virtual Try-On pipeline.

Thin Plate Spline provides smooth and mathematically stable geometric transformations, making it highly effective for initial garment adaptation.

The Geometric Matching Module improves fitting quality by learning complex relationships between body structures and clothing images.

HR-VITON extends these concepts by generating high-resolution outputs that preserve garment textures, folds, and fine details.

The research also highlights that successful Virtual Try-On systems require the integration of multiple AI components, including body segmentation, pose estimation, geometric matching, and image synthesis.

Proper handling of edge cases further improves system robustness and user satisfaction.

11. Learning Outcomes

During this research, the following concepts were studied and analysed:

- Fundamentals of Virtual Try-On systems.
- Principles of garment warping.
- Thin Plate Spline (TPS) transformation.
- Geometric Matching Module (GMM).
- HR-VITON architecture.
- Warp control point generation.
- Affine transformation techniques.
- Handling edge cases in Virtual Try-On applications.
- Applications of AI in fashion technology.
- Future directions of intelligent garment fitting systems.

This research provided a deeper understanding of how Artificial Intelligence combines computer vision and deep learning techniques to create realistic digital clothing experiences.

12. Conclusion

This research explored the Artificial Intelligence techniques used in modern Virtual Try-On systems, with particular emphasis on garment warping and geometric matching.

The study demonstrated how Thin Plate Spline transformation, Geometric Matching Modules, and HR-VITON architectures work together to generate realistic digital garment fitting experiences. It also highlighted the importance of body segmentation, landmark detection, and image synthesis in producing visually convincing results.

Furthermore, the research showed that handling difficult scenarios such as extreme body poses, occlusions, and complex backgrounds is essential for building reliable Virtual Try-On applications.

Overall, AI-driven garment warping technologies represent a major advancement in digital fashion and e-commerce. Continued improvements in deep learning, computer vision, and large-scale fashion datasets are expected to further enhance the realism, personalization, and effectiveness of future Virtual Try-On platforms.